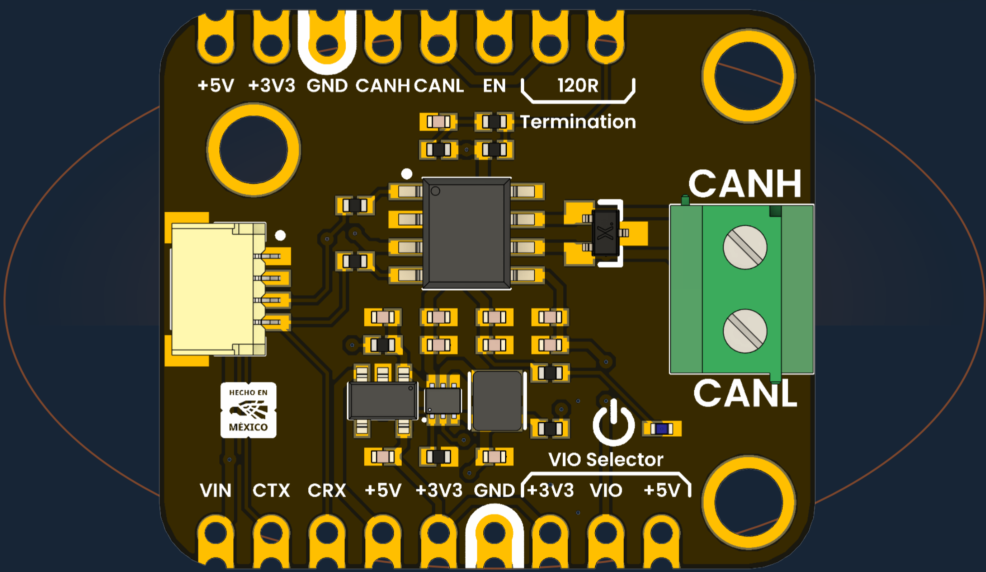
PRODUCT REFERENCE MANUAL

DEVLAB FACTOR

TCAN1051HVD CAN Transceiver Module

Mfr. Part #: UE0085

Version 1.0.0



Fault-protected high-speed CAN physical-layer interface with selectable 3.3 V or 5 V controller logic.

Classic CAN | CAN FD · 2 Mbps | 3.3 V / 5 V I/O | 120 Ω termination

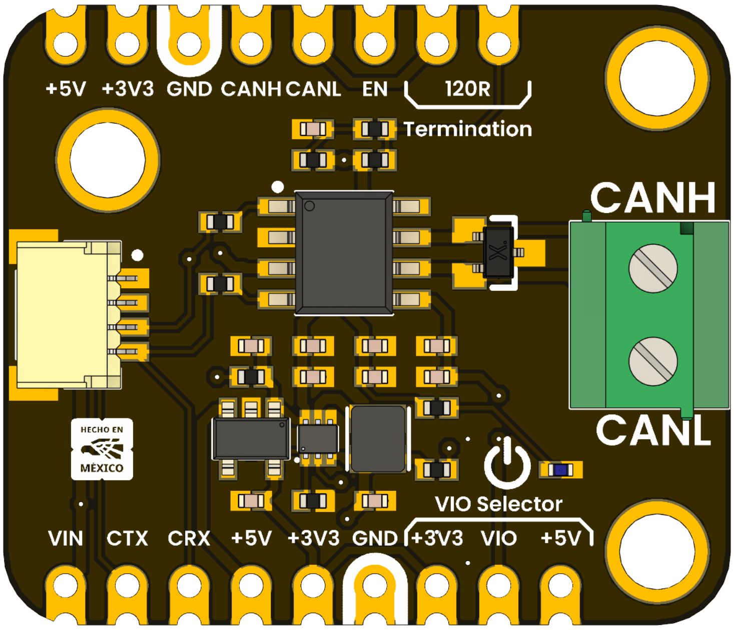
DOCUMENT MAP

Contents

Description

The **DevLab TCAN1051HVD CAN Transceiver Module** is a compact physical-layer interface for Classic CAN and CAN FD networks. It connects a microcontroller or stand-alone CAN controller to the differential CAN bus through the CTX (TXD), CRX (RXD), CANH, and CANL signals.

The module is built around the Texas Instruments TCAN1051HVD, a fault-protected high-speed CAN transceiver with a separate VIO supply. An onboard boost converter and 3.3 V regulator generate the transceiver and logic rails from VIN. The VIO selector allows direct use with either 3.3 V or 5 V controllers.



> This module implements the CAN physical layer only. The host must provide a CAN controller or a microcontroller with an integrated CAN/TWAI peripheral.

Applications

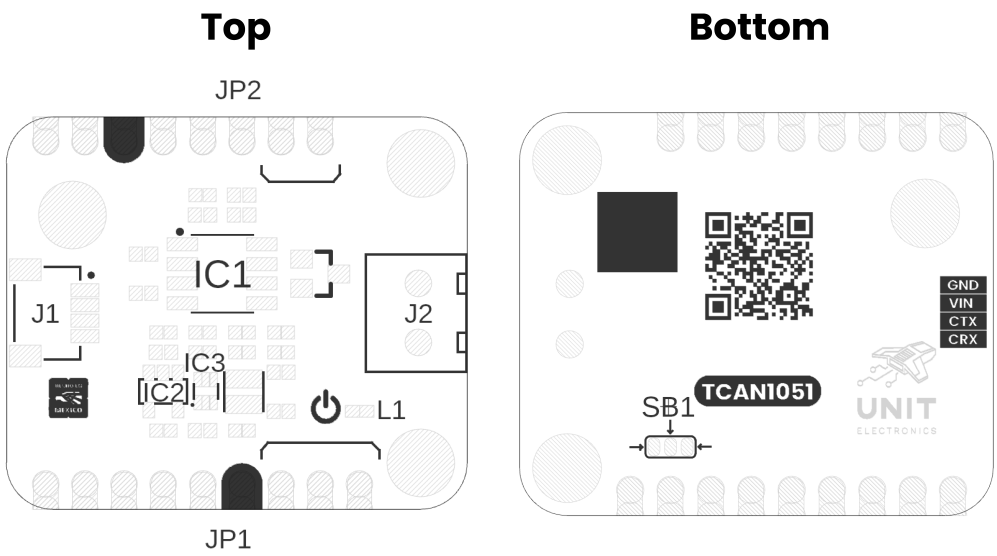
- Industrial automation and distributed control
- Robotics and mobile platforms
- Building and environmental control
- Automotive prototyping and diagnostics
- CANopen, DeviceNet, NMEA 2000, ISOBUS, and other CAN-based networks

Hardware Features

- TCAN1051HVD high-speed, fault-protected CAN transceiver
- Classic CAN and CAN FD operation up to 2 Mbps
- Selectable 3.3 V or 5 V logic interface (VIO)
- Onboard TPS61023 5 V boost converter and AP2112K 3.3 V regulator
- Switchable split 120 Ω termination (2 × 60 Ω with 4.7 nF center capacitor)
- PESD2CANFD24L-UX protection device on CANH and CANL
- 22 Ω series resistors on CTX and CRX
- 3.81 mm screw terminal for the CAN bus
- 4-pin, 1 mm-pitch connector for VIN, GND, CTX, and CRX
- Castellated headers for breadboard, carrier-board, or direct-wire integration
- Power indicator LED and exposed regulator enable signal

1 The Board

The module separates the controller-side logic interface from the differential CAN bus. Controller signals are available at the lower castellated header and the 4-pin connector, while CANH and CANL are available at the upper header and the screw terminal.



Views of Board Topology

1.1 Package Contents

- 1 × DevLab TCAN1051HVD CAN Transceiver Module (UE0085)

Headers, mating cables, and bus wiring are not included unless stated by the product listing.

1.2 Main Components

Reference	Component	Function
IC1	TCAN1051HVD	CAN physical-layer transceiver
IC2	AP2112K-3.3	3.3 V linear regulator
IC3	TPS61023	5 V boost converter
D1	PESD2CANFD24L-UX	CANH/CANL transient protection
D2	Green LED	Power indication
J1	4-pin, 1 mm-pitch connector	VIN, GND, CTX, and CRX interface
J2	2-position, 3.81 mm terminal	CANH and CANL bus connection
JP1	Termination selector	Connects the onboard split 120 Ω termination
SB1 / JP4	VIO selector	Selects 3.3 V or 5 V logic levels

2 Ratings and Electrical Characteristics

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VIN	Module input voltage	3.3	—	5.0	V
VCC	TCAN1051HVD bus supply	4.5	5.0	5.5	V
VIO	Selected I/O level-shifting supply	2.8	3.3 / 5.0	5.5	V
IO(RXD)	Recommended CRX output-current range	−2	—	2	mA
TA	IC operating free-air temperature	−55	—	125	°C
CAN FD	Supported data rate for the TCAN1051HVD variant	—	—	2	Mbps

VIN is the module-level input recommendation. VCC, VIO, I/O current, temperature, and CAN FD values are TCAN1051HVD specifications. The complete assembled module has not been independently qualified over the full IC temperature range.

2.2 Selected Transceiver Characteristics

Unless noted otherwise, values below are specified by Texas Instruments over the recommended operating conditions.

Symbol	Description	Min	Typ	Max	Unit
VIH	CTX high-level input voltage	0.7 × VIO	—	—	V
VIL	CTX low-level input voltage	—	—	0.3 × VIO	V
VOH	CRX high-level output voltage at −2 mA	0.8 × VIO	—	—	V
VOL	CRX low-level output voltage at 2 mA	—	—	0.2 × VIO	V
VO(DOM), CANH	Dominant CANH output voltage	2.75	—	4.5	V
VO(DOM), CANL	Dominant CANL output voltage	0.5	—	2.25	V

Symbol	Description	Min	Typ	Max	Unit
VOD(DOM)	Dominant differential output, 50–65 Ω load	1.5	—	3.0	V
VCM	Receiver common-mode range	−30	—	30	V
tPROP(LOOP)	Total loop delay	—	100 / 110	160 / 175	ns

2.3 Absolute Maximum Ratings

Absolute maximum ratings are stress limits, not normal operating conditions.

Symbol	Description	Min	Max	Unit
VCC	TCAN1051HVD 5 V supply	−0.3	7	V
VIO	TCAN1051HVD I/O supply	−0.3	7	V
VBUS	CANH or CANL, H-suffix device	−70	70	V
VDIFF	CANH-to-CANL differential voltage, H-suffix device	−70	70	V
IO(RXD)	CRX output current	−8	8	mA
TJ	IC junction temperature	−55	150	°C
TSTG	IC storage temperature	−65	150	°C

Do not apply the ±70 V bus-fault rating to VIN, CTX, CRX, EN, VIO, 3V3, or 5V.

3 Functional Overview

3.1 Signal Path

CTX drives the TCAN1051HVD TXD input. A logic LOW requests a dominant bus state, and a logic HIGH requests a recessive state. The transceiver converts this logic signal into the CANH/CANL differential output.

CRX is driven by the TCAN1051HVD RXD output. CRX is LOW when the receiver detects a dominant bus state and HIGH when it detects a recessive state. The CRX output level follows the selected VIO rail.

The TCAN1051HVD includes dominant-timeout, thermal-shutdown, and undervoltage protection. The bus and logic pins present high impedance when the transceiver is unpowered.

3.2 Power Architecture

VIN feeds the TPS61023 boost converter, which generates the 5 V transceiver supply. The AP2112K then generates the 3.3 V rail. Both regulated rails are exposed on the castellated headers.

The VIO selector connects the transceiver I/O supply to either 3.3 V or 5 V:

Host logic	VIO selector
3.3 V MCU or CAN controller	3V3
5 V MCU or CAN controller	5V

Select exactly one side of the VIO solder selector. Bridging both sides shorts the 3.3 V and 5 V rails and can damage the module or host.

EN controls the TPS61023 regulator. The onboard bias enables the regulator for normal use; driving EN LOW disables the generated supply rails and transceiver.

3.3 Bus Termination

The onboard termination is a split 120 Ω network made from two 60 Ω resistors. Its midpoint is AC-coupled to ground by 4.7 nF to reduce common-mode noise.

Close the **120R Termination** selector only when this module is installed at one physical end of the CAN trunk. A correctly wired point-to-point bus normally has one 120 Ω terminator at each end (approximately 60 Ω measured between CANH and CANL with power removed). Leave the selector open at intermediate nodes.

3.4 Bus Protection

CANH and CANL are protected by a PESD2CANFD24L-UX device. The TCAN1051HVD H-suffix transceiver provides a ±70 V bus-fault rating and a ±30 V receiver common-mode range. These protections improve robustness but do not replace correct grounding, cabling, termination, or system-level surge design.

4 Connectors and Pinouts

4.1 Signal Reference

Label	Direction	Description
VIN	Power input	Module input supply, 3.3 V to 5.0 V
GND	Power	Common module and bus reference
CTX	Input	Controller transmit signal; LOW = dominant
CRX	Output	Controller receive signal; LOW = dominant
CANH	Bus I/O	CAN high bus line
CANL	Bus I/O	CAN low bus line
EN	Input	5 V boost-converter enable; LOW disables the module
5V	Power output	Generated 5 V transceiver rail
3V3	Power output	Generated 3.3 V logic rail
VIO	Power	Selected transceiver logic supply (3.3 V or 5 V)

4.2 J1 Controller Connector

J1 is a 4-position, 1 mm-pitch connector. Use the bottom-side silkscreen to confirm pin order before connecting a cable.

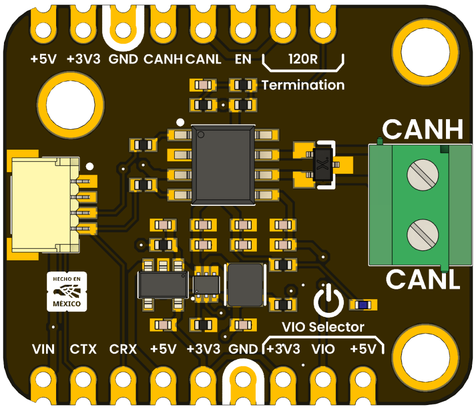
Signal	Connection
GND	Host ground
VIN	Host power supply
CTX	Host CAN TX output
CRX	Host CAN RX input

4.3 J2 CAN Bus Terminal

Position	Signal	Connection
Upper	CANH	CAN bus high line
Lower	CANL	CAN bus low line

4.4 Castellated Headers

The castellated pads duplicate the controller, bus, regulator, and control signals for solderless prototyping or carrier-board mounting. Their labels are printed directly on the top silkscreen.


+5V
+3V3
GND
CANH
CANL
EN
120R
Termination
CANH
CANL
VIO Selector
VIN
CTX
CRX
+5V
+3V3
GND
+3V3
VIO
+5V

5 Board Operation

5.1 Before Connecting the Module

1. Confirm that the host includes a CAN controller or CAN-capable peripheral.
2. With power removed, set VIO to match the host logic voltage (3.3 V or 5 V).
3. Enable the onboard 120 Ω termination only if this node is at an end of the CAN trunk.
4. Check CANH/CANL polarity and provide a common ground where required by the system design.

5.2 Basic Connection

Module	Host or bus
VIN	3.3 V to 5.0 V supply
GND	Host/bus reference ground
CTX	CAN controller TX output
CRX	CAN controller RX input
CANH	CANH trunk conductor
CANL	CANL trunk conductor

Keep the CANH/CANL pair twisted, minimize stubs, and place termination only at the two physical ends of the main bus.

5.3 Power-Up Check

After applying VIN:

- The green power LED should illuminate.
- The 5V rail should measure approximately 5 V.
- The 3V3 rail should measure approximately 3.3 V.
- CRX should idle HIGH at the selected VIO level when the bus is recessive.
- CANH and CANL should both be near the recessive common-mode level when idle.

Do not continue if either regulator rail is shorted, if the device becomes hot, or if the VIO selector connects both rails.

5.4 Controller Configuration

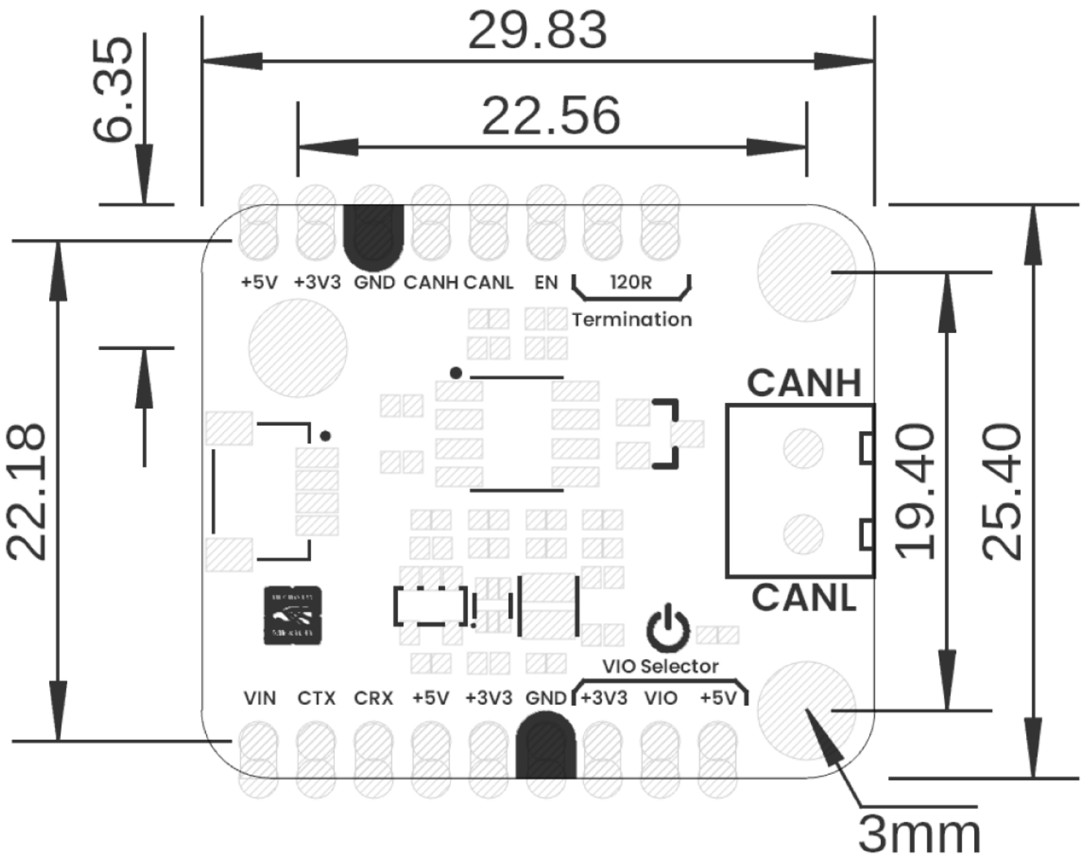
Configure the host CAN controller with the same nominal bit rate as every other node. CAN FD data-phase timing must also match when CAN FD is used. The TCAN1051HVD supports Classic CAN and CAN FD data rates up to 2 Mbps; protocol timing, message identifiers, filters, and acknowledgments are handled by the host controller and software.

5.5 Troubleshooting

Symptom	Checks
No power LED	VIN polarity/level, GND continuity, EN state
Host receives nothing	VIO setting, CRX pin mapping, bit rate, CANH/CANL polarity
Bus errors or missing ACK	Second active node, matching bit rate, two end terminators
Communication degrades with cable length	Twisted pair, stub length, topology, termination, bit rate
Bus held dominant	CTX stuck LOW, incorrect host pin mode, wiring short

6 Mechanical Information

The nominal board outline is **29.83 mm × 25.40 mm**. The drawing below also identifies the 3 mm mounting holes and principal spacing dimensions. All values are in millimeters.



Board Dimensions in Millimeters

Use the released mechanical drawing or a physical sample when designing mating enclosures or production fixtures; the rendered image is not a 1:1 fabrication template.

6.1 Board Views

<table>
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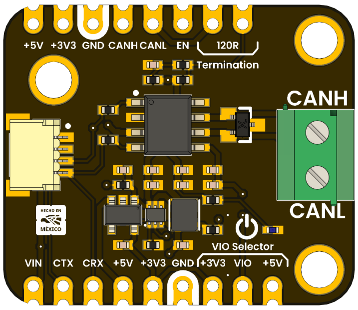

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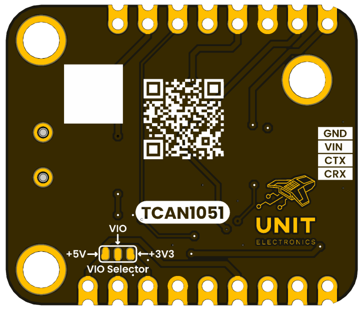

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7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Reference	Link
Product repository	UNIT-Electronics-MX/ unit_devlab_tcan1051hvd_can_transceiver_module
Arduino library	UNIT-Electronics-MX/UE0085-TCAN1051HVD
TCAN1051HV datasheet	Texas Instruments SLLSES8E
ISO 11898-2 overview	Refer to ISO 11898-2:2016 for the high-speed CAN physical layer

The released schematic and the manufacturer datasheet are also included in the repository under `hardware/` and `hardware/resources/datasheet/`.

9 Appendix

9.1 Schematic

The complete revision 1 schematic is distributed with this product reference:

Open the UE0085 TCAN1051HVD module schematic

9.2 Design Notes

- R1 provides the switchable 120 Ω termination.
- R2 and R3 form the 2 × 60 Ω split termination.
- C5 (4.7 nF) AC-couples the split-termination midpoint to ground.
- R4 and R5 (22 Ω) are in series with CTX and CRX.
- D1 protects CANH and CANL.
- The S (silent-mode) pin is fixed for normal transceiver operation on this module and is not exposed as a user control.